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title: "Querying_and_Manipulation.sql"
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Title: CS 504 project stage 2 Querying_and_Manipulation.sql
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```

-- question 1

-- List the names and joined dates of all members who registered in the year 2018.

```
select mt.join_date, mt."name"
from
  member_tbl mt
where mt.join_date between date '2018-01-01' and date '2018-12-31';
```

-- question 2

-- Retrieve the names of members along with the total count of books they have borrowed.

```
select
  mt.name,
  count(b.material_id ) as total_books_borrowed
from
  member_tbl mt
left join
  borrow b ON mt.member_id = b.member_id
group by mt.member_id, mt."name";
```

-- question 3

-- Calculate the average borrowing duration (in days) for all borrowed books.

-- this one is tricky.....

```
select
  (b.return_date - b.borrow_date )/count(b.borrow_date) as average_borrow_time
from
  borrow b
group by b.return_date, b.borrow_date

select
  b.material_id,
  count(b.material_id) is distinct
  (b.return_date - b.borrow_date ) as average_borrow_time,
from
  borrow b
group by b.return_date, b.borrow_date, b.material_id
```

-- sum (return - borrow) / number of times borrowed

```
select count(b.material_id) distinct
from borrow b;
```

```
select b.borrow_id
from borrow b
where b.material_id is
```

-- question 4

-- Identify the author who has the highest number of books in the library, including their name

-- and the total book count.

```
select
  au.name,
```

```
    count(distinct a.material_id) as total_book_count
from authors au
join authorship a
  on au.author_id = a.author_id
group by au.author_id, au.name
order by total_book_count desc
limit 1;
```

```
-- question 5
-- List all authors along with the titles of the books they have written.
select
  a.name as author,
  m.title as book_title
from authors a
left join authorship au
  on au.author_id = a.author_id
left join material m
  on m.material_id = au.material_id
order by a.name, m.title;
```

```
-- question 6
-- Determine the book that has been borrowed the most times.
select
  m.title,
  count(*) as times_borrowed
from borrow b
join material m
  on m.material_id = b.material_id
group by m.material_id, m.title
order by times_borrowed desc
fetch first 1 row with ties;
```

```
-- question 7
-- Retrieve the list of members who have borrowed books but have not yet returned them
-- (return_date is NULL).
select
  mtb.name
from member_tbl mtb
join borrow b
  on b.member_id = mtb.member_id
where b.return_date is null;
```

-- The check below shows inconsistent behavior....

```
select
  count(mtb."name")
from member_tbl mtb
join borrow b
  on b.member_id = mtb.member_id
where b.return_date is null;
-- returns 28
```

```
select
  count(member_id)
from borrow
where return_date is null;
-- returns 22
```

```
select
  count(mtb."name")
from member_tbl mtb
join borrow b
  on b.member_id = mtb.member_id
where b.return_date is null;
```

```
-- returns 22, correct solution!
```

```
-- question 8
-- Identify the staff member who has processed the highest number of book borrow
transactions.
select
    s.name,
    count(b.borrow_id) as transactions_processed
from staff s
join borrow b
    on b.staff_id = s.staff_id
group by s.staff_id, s.name
order by transactions_processed desc
limit 1;
```

```
-- question 9
-- Find the most borrowed book genre based on borrowing records.
select
    g.name as genre,
    count(*) as times_borrowed
from borrow b
join material m
    on m.material_id = b.material_id
join genre g
    on g.genre_id = m.genre_id
group by g.genre_id, g.name
order by times_borrowed desc
limit 1;
```

```
-- question 10
-- List all books that are overdue (i.e., where the due date has passed but the return_date
is
-- still NULL).
select
    m.title
from material m
join borrow b
    on m.material_id = b.material_id
where b.return_date is null
    and b.due_date < current_date;
```

```
...
```